

Project Summary: Multigraded Homological Algebra & Geometry

Overview.

Syzygies, the relations among equations, encode algebra and geometry, but much of the classical theory applies works best for standard graded rings. Rings with weighted or multiple gradings are unavoidable in modern commutative algebra and geometry: modular curves, section rings of stacky curves, and Cox rings of toric varieties all require multi/nonstandard gradings. This project develops multigraded homological algebra in three directions: (i) the syzygies of stacky curves, aimed at an analog of Green’s conjecture for stacky curves; (ii) the shape and bounds of multigraded Castelnuovo–Mumford regularity and its connections to Betti numbers and toric geometry; and (iii) a regularity theory on the Cox category that repairs fragilities of multigraded regularity. New *Macaulay2* tools, mentoring, educational resources, and organizing complement the research.

Intellectual Merit.

The first direction develops syzygies of stacky curves, where fractional divisors play the role classically played by degree. The PI will determine the generator degrees of stacky canonical rings, sharpening Voight and Zureick-Brown’s stacky Noether–Petri theorem, and relate their syzygies to those of the coarse curve. Since stacky points enlarge the Betti table beyond its classical shape, the stacky analog of Green’s conjecture splits in two: a weak form, predicting that the coarse curve controls the linear strand and weighted N_p -properties, and a strong form, predicting that the full Betti table depends only on the coarse curve and the signature. Further projects study high-degree line bundles and log canonical rings on stacky curves.

The second direction explains the shape of multigraded regularity on smooth projective toric varieties, where regularity is a region the Betti table does not determine. The PI proposes that support geometry governs the directions in which this region opens and that syzygies govern where its boundary lies. Together, these ideas will clarify how regularity relates to Betti numbers and yield the first finite algorithm for computing a module’s full regularity region. A central goal is a toric analog of the Gruson–Lazarsfeld–Peskine theorem, which bounds the regularity of a projective curve by its degree: the PI conjectures an explicit bound in terms of the curve’s numerical class and combinatorial data of the ambient variety. Finally, the PI will seek combinatorial formulas for the regularity of torus-invariant subvarieties, which extend to bounds on arbitrary subvarieties.

The third direction asks whether multigraded regularity is the right definition in the wrong category. Working in the Cox category, which unifies all toric birational models sharing a Cox ring, the PI uses the uniform Hanlon–Hicks–Lazarev (HHL) resolution and the Bondal–Thomsen collection to propose a new definition of regularity that recovers the classical one on projective space and makes the structure sheaf $\mathbf{0}$ -regular on every smooth projective model. Central questions in this project are whether regularity in the PI’s sense persists under nef twists, whether it implies global generation, and how it relates to MacLagan–Smith regularity.

Two software projects yield community tools: further development of the *Macaulay2* package `toricStacks` and a database of the expert-chosen examples in *Macaulay2*’s documentation, tracking authorship, as a first step toward community-centered AI/ML tools.

Broader Impacts.

The PI advises three Ph.D. students on problems central to the proposal and will continue mentoring undergraduates. The PI will turn her Preliminary Arizona Winter School lectures and commutative algebra course notes into a modern, inquiry-based introductory book on algebraic geometry and commutative algebra. The PI is leading a `toricStacks` group at the 2027 *Macaulay2* conference and will host *Gems of Commutative Algebra III* in Fall 2028. The PI has submitted a proposal to organize a 2030 Fields thematic semester in combinatorial algebraic geometry. The PI will continue organizing *Unwritten Rules* and working with Dartmouth’s AWM chapter. The PI’s work on AMS committees and the *Spectra* Board of Trustees extends these efforts nationally.